Campus Gym Buddy

Systems Analysis & Design

System Proposal

[**INSTRUCTIONS**:

- Complete each section below.
- Replace all text inside square brackets [] with your writing
- Remove all instructions.
- Use a spell checker like Grammarly to revise the grammar of your writing.]

Executive Summary (Full System)
Campus Gym Buddy / Workout Group System is a student-focused platform created to help verified college students find workout partners and small fitness groups on campus. Many students want to go to the campus gym, but they often face challenges such as feeling awkward working out alone, lacking confidence as beginners, having trouble finding people with similar fitness goals, and losing motivation without support from others. This system is designed to solve those problems by offering a safe and organized space where students can build profiles, join or create workout groups, and schedule gym sessions with fellow students. What makes this system different is that it is centered on campus-based social fitness rather than workout tracking. Instead of focusing on calories, repetitions, or performance analytics, it highlights connection, accountability, and consistency. Only verified students can use the platform, and workout groups are limited to five members to make planning easier and more personal interactions. Students can also match with others based on their fitness goals, experience level, and availability, which helps them find workout partners who are a good fit. For the prototype, the system will mainly concentrate on essential data management features such as managing student profiles, creating workout groups, saving session information, and monitoring group membership. Together, these functions support the overall goal of the system, which is to help students find a safe, structured, and encouraging way to stay active with other students on campus.

System Request (Full System)	
Project sponsor(s)	Brian Barrios, Md Usman Goni, Pablo Jacme
Business need	Many college students can use a campus gym, but simply having access does not mean they will actually go. One major issue is that students often feel uncomfortable working out alone, especially if they are beginners or do not know how to use gym equipment or follow workout routines. Some students also want a workout partner or group for motivation, safety, accountability, or social support, but they do not have an easy way to find others with similar fitness goals, schedules, or experience levels.

	Although campuses may provide gym facilities, they often do not offer a structured system that helps students connect specifically for workout sessions. General social media platforms and messaging apps are not built for this purpose, and they usually lack features such as campus verification, group-size limits, and organized session planning. Because of this, many students may avoid going to the gym, going irregularly, or struggling to build consistent exercise habits due to a lack of support and structure. The Campus Gym Buddy / Workout Group System is needed to address this problem by giving verified students a dedicated platform where they can create and join small workout groups, match with others based on fitness goals and skill level and organize gym sessions in a more structured and supportive way.
Business value	The proposed system provides both tangible and intangible values. **Tangible value** • Increases student participation in campus gym activities • Encourages more consistent use of existing campus fitness resources • Provides an organized way to manage workout groups and scheduled sessions • Reduces the need for informal coordination through scattered texts or social media messages **Intangible value** • Helps students feel more comfortable and supported when using the gym • Improves motivation and accountability through peer interaction • Creates a stronger sense of campus community • Supports student wellness, confidence, and belonging • Provides a safer and more trustworthy experience by limiting access to verified students Overall, the system adds value by transforming the gym from an individual activity into a more accessible and collaborative campus experience.

Business Requirements (Prototype)

Student profile management

- The system must store student profile information for each user.
- The system must store each student's name, student email, skill level, and workout goals.
- The system must indicate whether a user is a verified student.

- The system must allow a student profile to be associated with one or more workout groups.

Workout group management

- The system must allow the creation of workout groups.
- The system must store a group name, workout focus, skill level, and maximum capacity.
- The system must limit each workout group to no more than five students.
- The system must allow a workout group to be identified as open to joining or full.
- The system must associate each workout group with the student who created it.

Group membership management

- The system must record which students have joined which workout groups.
- The system must allow a student to belong to multiple workout groups, if permitted by the business rules.
- The system must track the date a student joins a group.
- The system must allow the system to determine the current number of members in a group.

Gym session management

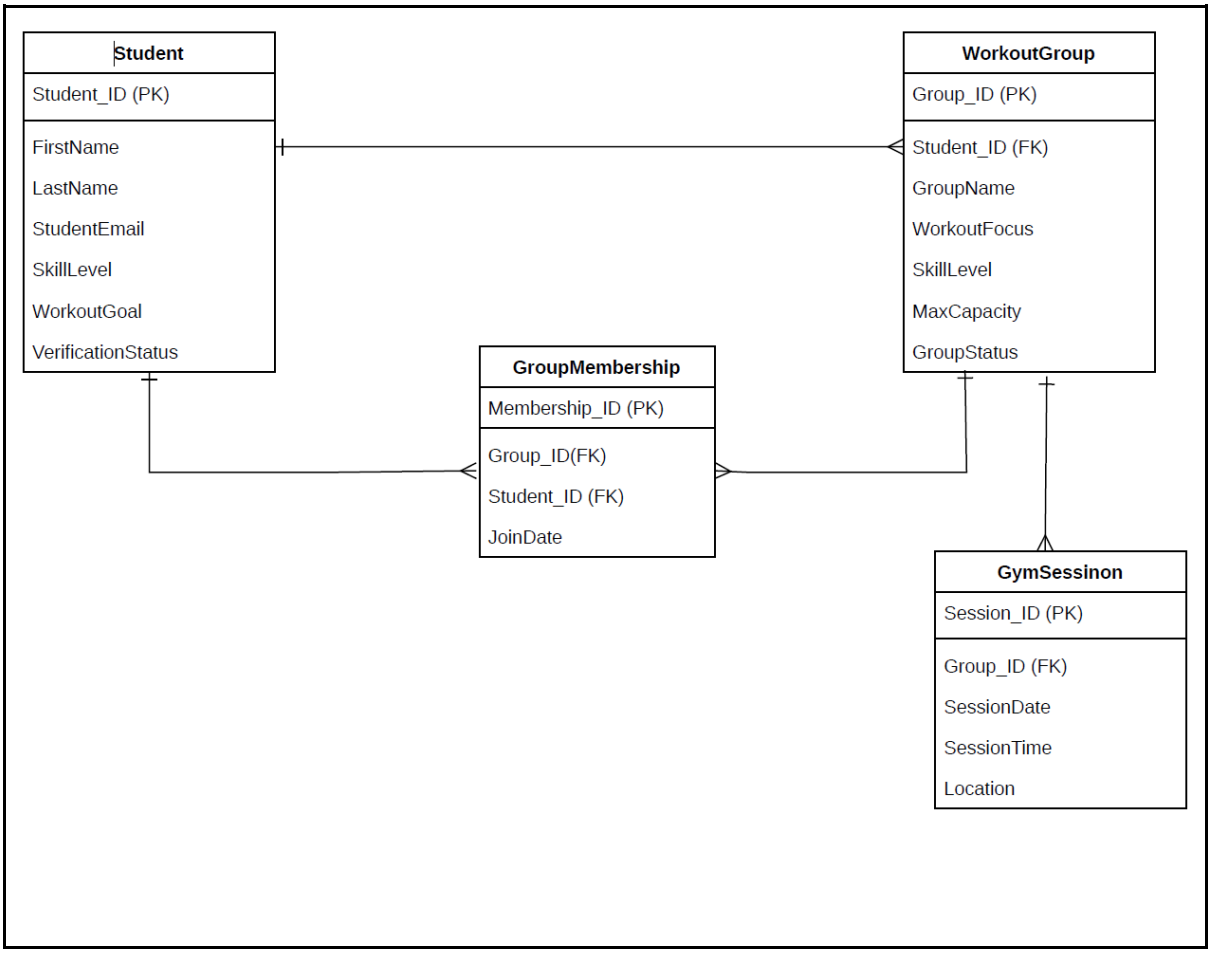
- The system must store planned gym sessions for workout groups.
- The system must store the date, time, and location of each gym session.
- The system must associate each gym session with one workout group.
- The system must allow a workout group to have multiple gym sessions over time.

Search and matching support

- The system must support identifying workout groups by skill level.
- The system must support identifying workout groups by workout goals or focus.
- The system must support identifying available groups that have not reached maximum capacity.

Data Modeling (Prototype)

Link to analysis models: https://drive.google.com/file/d/1-V-gB00NgW5S14zEFjhq08-egPfdDy5X/view?usp=sharing

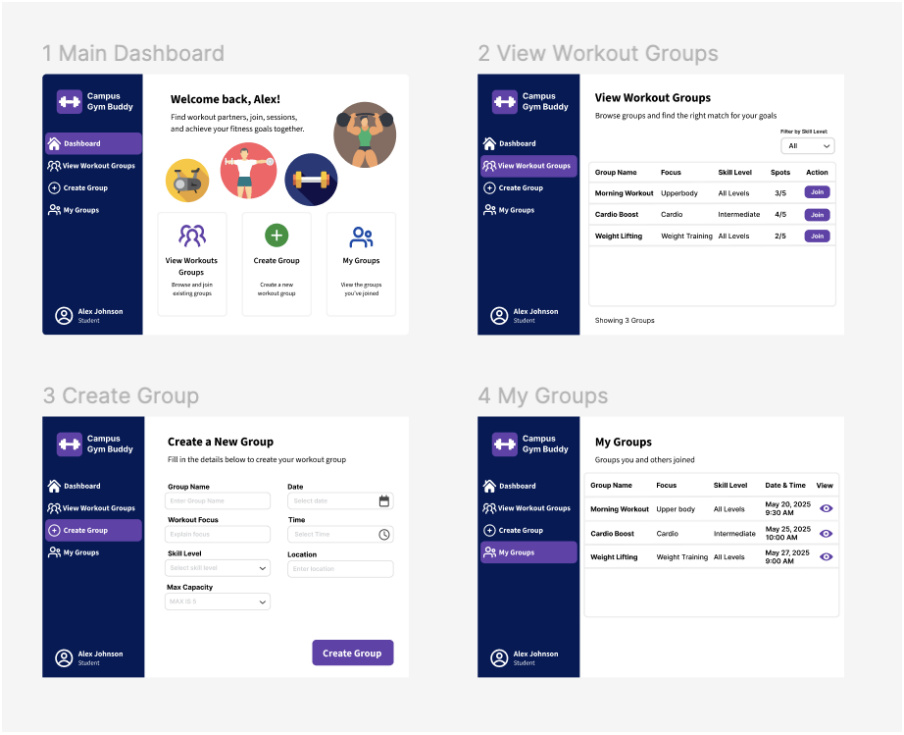


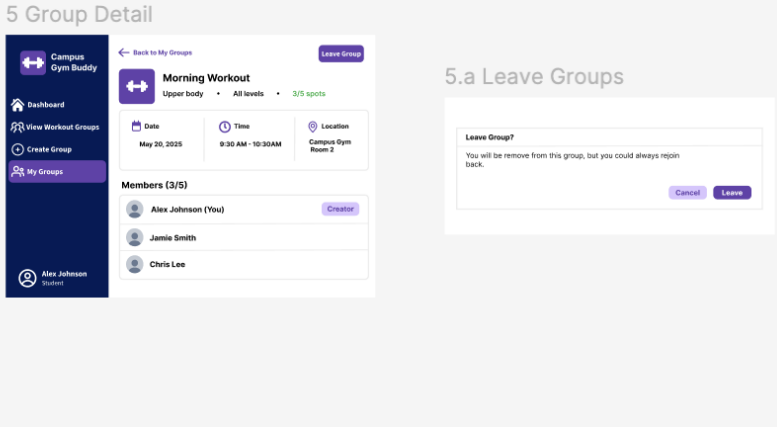
Prototype

Link to prototype: If using Figma, include the URL to your prototype. Ensure your URL is public.]
Figma work
https://www.figma.com/design/cQJASUKAZQzINnzF0PueTR/Campus-Gym-Buddy?node-id=2-3&t=EQOgRY5M6SqWwu4s-1

Figma Ready Demo
https://www.figma.com/proto/cQJASUKAZQzINnzF0PueTR/Campus-Gym-Buddy?node-id=249-19&p=f&t=eWd4TCyoj4htv1XR-1&scaling=contain&content-scaling=fixed&page-id=2%3A3
The interfaces in your prototype should be connected.

- One interface should be the primary interface (the starting point or dashboard).
- The other interfaces should be accessible from the main interface, demonstrating user interactions.





This iteration focuses on gathering feedback from your instructor.

Use realistic placeholder content, such as meaningful text in headers and body sections, and interface elements like logos and icons.]

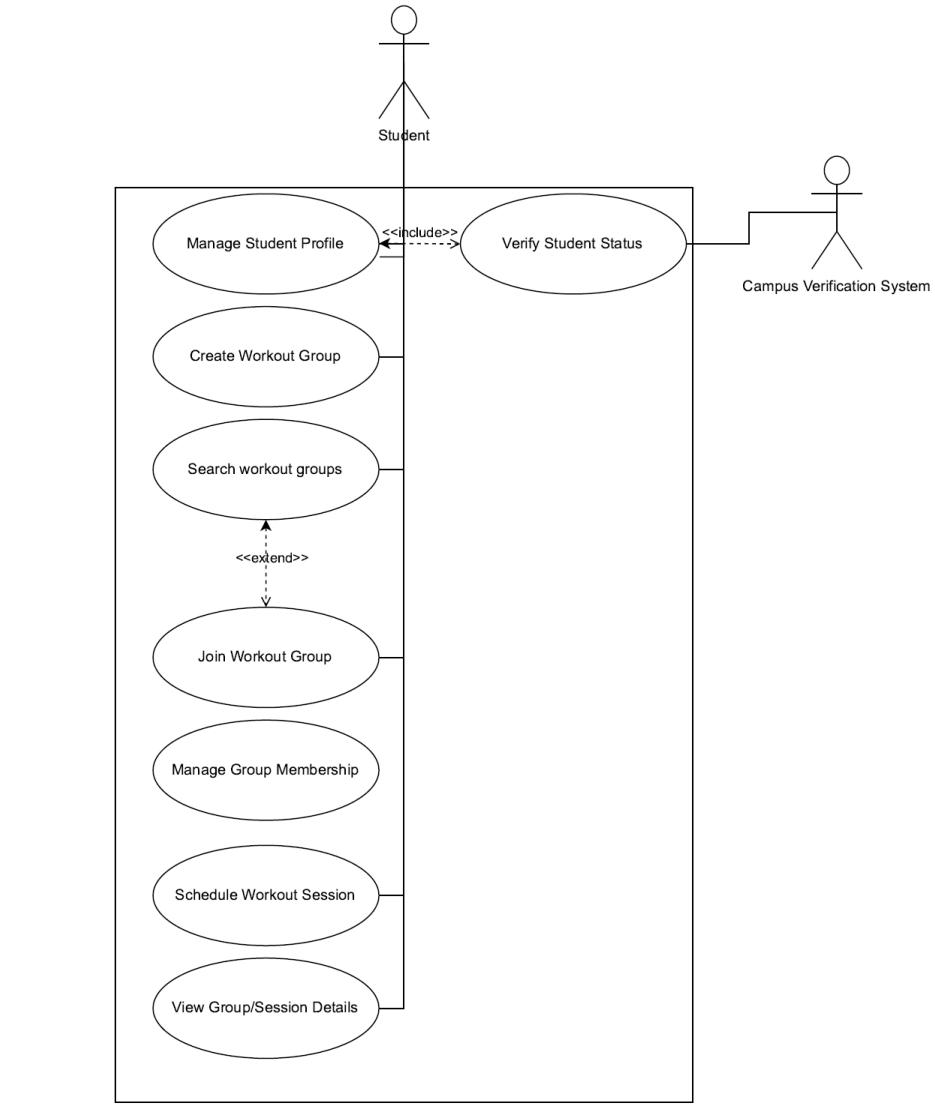
Functional Modeling (Prototype)

Link to analysis models: [URL to analysis models]

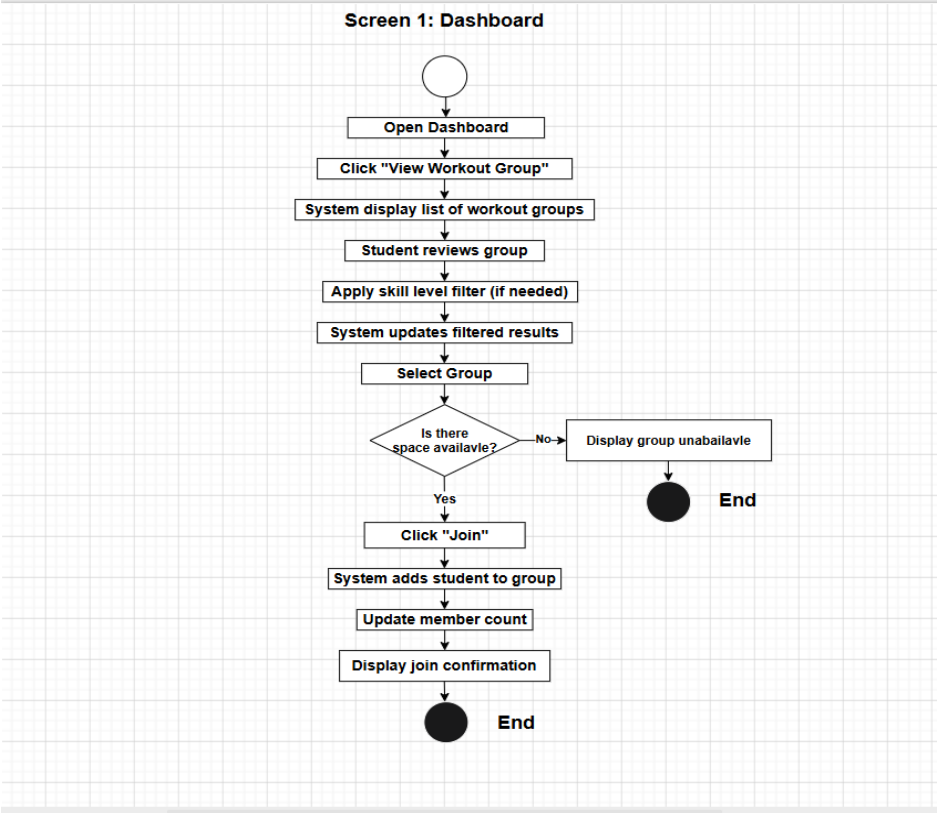
[Using Draw.io, create the following diagrams to document the functionality represented in your prototype. These diagrams should align with the features and processes you have modeled in your prototype and help demonstrate how users will interact with it.

- **One use-case diagram** that provides a high-level overview of the prototype's functionality, showing the main actors and their interactions with the system.

https://drive.google.com/file/d/1IU5uXTP4rKL28v8EqKhaCoXN0x7FzgDf/view?usp=sharing



- **A set of activity diagrams** that illustrate key processes or particular tasks supported by the prototype. Each diagram should clearly depict the flow of activities involved in carrying out specific user tasks included in your prototype.
  https://drive.google.com/file/d/168iCf9WU_9vKezLbk7SPvHHp_Wcf81Ko/view?usp=sharing



Please include a link to your diagrams in the space provided above. Ensure that all diagrams follow the notation standards covered in class.]

References:

- Centers for Disease Control and Prevention. (2025). *Promoting physical activity in college-aged students*. https://www.cdc.gov/pcd/issues/2025/25_0118.htm

- Smith, J., & Johnson, L. (2024). The impact of peer-led fitness programs on student well-being. *Journal of American College Health*. https://journals.sagepub.com/doi/full/10.1177/15588661241257478
- WorldMetrics. (2026). *College students exercise statistics*. https://worldmetrics.org/college-students-exercise-statistics/
- Zhang, Y., Chen, X., & Liu, H. (2022). Physical activity and fitness among university students: Barriers and influencing factors. *International Journal of Environmental Research and Public Health*. https://pmc.ncbi.nlm.nih.gov/articles/PMC8750240/